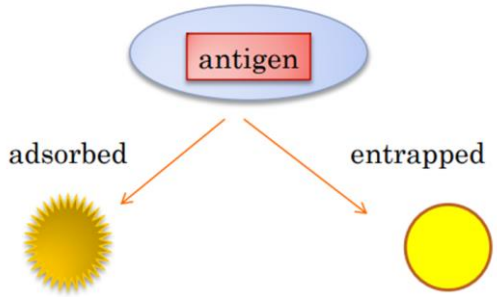
91. Nanovaccine

- **Introduction**: A vaccine is a biological preparation that improves immunity to a particular disease.
- A vaccine typically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microbe or its toxins.
- The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and "remember" it, so that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters.
- **Nanovaccine**: Nano is very vast field and it can be applied to any area, one of such area is vaccine.
- It provides a different route of administration of vaccine.
- Nano vaccine can be designed, manufactured and introduced into the human body to improve health, including cellular repairs at the molecular level.
- The nanomaterial is so small that it can easily enter the cell; therefore, nanomaterials can be used in vivo or in vitro for biological applications.
- Delivery of Nanovaccine: Following are the different way to deliver vaccine at nano level
- **Nanobead:** Inert solid bead Size range 20-200nm.
- When antigens are adsorbed on the surface of bead it has been shown to stimulate CD8 -T cell response.
- The size of the bead play a major role in eliciting a combined response of humoral and cell-mediated immunity.
- Antigen covalently linked to inert nano-beads with a size of ~50 nm is preferentially taken up by dendritic cells, thus inducing humoral as well as cell-mediated immune responses.
- **2. Polymeric nanoparticles**
- Biodegradable, biocompatible polymers have been approved for use in humans.
- Poly(D, L-lactide-co-glycoside) (PLG) and polylactide (PLA).



- Polyglycolide (PLG) have been extensively used to encapsulate antigens.
- PLG forms lactic and glycolic acids, after hydrolysis of α-hydroxyl acids, yielding small spherical polymeric particles 1–100 nm in size.
- Adsorbed antigen offers improved stability and activity over encapsulated antigen by avoiding exposure to organic solvents used during formulation and acidic pH conditions caused by degradation of the polymer.
- 3. **Nano-emulsion**
- Size of globule (100-400nm)
- Nano-emulsion vaccine does not require refrigeration and is stable for 6 months.
- Nano-emulsion is non-toxic, pain free and avoids the risk of spreading needle-borne infections.
- Nano-emulsion of hepatitis B antigen, has been reported to be a safe and effective hepatitis B vaccine.

92. Nanovaccine II

- **Adjuvants:** These substances used in combination with a specific antigen that produced a more robust immunity than the antigen alone.
- It improves the immune response to vaccine antigen by different way:

1. Increasing immunogenicity of weak antigen.
2. Enhancing the speed and duration of the immune response.
3. Modulating antibody avidity, specificity, isotope or subclass distribution.
4. Examples
5. MF59
6. Monophosphoryl lipid A
7. Montanide™
8. Calcium phosphate nanoparticles.
9. Immunostimulant complexes
10. Cholesterol-bearing hydrophobized pullulan
11. Nanoparticles (CHP)

MF59

- MF59 is the only nano-sized vaccine adjuvant approved for human use.

- MF59 is an oil-in-water emulsion (≤ 250 nm droplets)
- It showed a 34-fold increase in antibody titers when immunized with glycoprotein D of herpes simplex virus (HSV) in guinea pigs.
- The mechanism of adjuvanticity of MF59 is believed to be through direct stimulation of cytokine production.

Monophosphoryl lipid A (MPL®)

- Monophosphoryl lipid A (MPL®) is an immunostimulant TLR-4 receptor agonist composed of detoxified lipopolysaccharide (LPS) from Salmonella Minnesota R595.
- It is a versatile vaccine adjuvant that may be included in aqueous formulations or in an oil-in water emulsion for a more dynamic response.

Montanide:

- There are several different types of Montanide™, including ISA 50V, 51, 206 and 720.
- ISA 50V, 51 and 720 are water-in-oil emulsions while ISA 206 is a oil-in-water emulsion.
- ISA 206 and 50V only used in veterinary vaccine formulations.
- SA 51 & ISA 720 are under investigation for use in humans.
- Emulsions of Montanide™ ISA 51 and 720 are composed of a metabolizable squalene-based oil with a mannide monooleate emulsifier.
- The immune enhancement produced by the Montanide™ emulsions is believed to be due to the formation of a depot at the site of injection.
- The emulsion vaccines against malaria, HIV and various cancer have been in phase I and/or II clinical trials.
- A phase I trial of a trivalent malaria vaccine containing ISA 720 induced both humoral and cellular immune responses

Calcium Phosphate NPs:

- Calcium phosphate nanoparticles are less than 1000nm in diameter.
- Nanoparticles can be generated by combining (while stirring) calcium chloride, sodium phosphate and sodium citrate.
- It produces longer duration of response as compare to Al salt.
- Phase I study showed that CaP is safe and non-toxic when administered subcutaneously.
- Vaccines utilizing CaP in preclinical studies include anthrax, HBV, flu (H5N1 avian and seasonal) and HSV-2.

Advantages of Nanovaccine

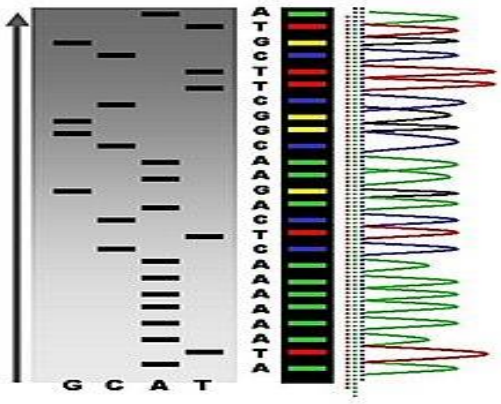
- Nano-vaccine have potential to deliver safe and more effective vaccine.
- Nanobead covalently coupled with antigen offer distinct advantages – a low dose of antigen is required, efficient processing by antigen-presenting cells and stability during storage.
- Encapsulated nanoparticles easily deliver antigen, protects the antigen from degradation and is found to be effective with a single dose due to slow release of the antigen.
- Many of the nano-vaccines are non-invasive, delivered by the oral or nasal route, diffusion patches or microneedle arrays, thus allowing pain-free delivery with minimal damage. This is an advantage over conventional vaccines, which are usually multi-injection, multi-dose delivery systems.

Disadvantages:

- Cost of production.
- Nanomaterials can change size, shape but not composition, which may change their toxicity.
- Small nanoparticles are cleared quickly from the body, large counterparts may accumulate in vital organs causing toxic problems.

93. Nanopore sequencing

- **DNA Sequencing**: DNA sequencing is the process of determining the precise order of the nucleotides within a DNA molecule
- DNA sequencing may be used to determine the sequence of individual genes, larger genetic regions, full chromosome or entire whole genome.



What is Nanopore?

- Nanopore = Very small hole
- Electrical current flows through the hole
- Introduce analyte of interest in the hole

- Identify analyte by the disruption or block to the electric current.

Introduction:

- The fourth-generation DNA sequencing technology
- Studies the interaction between DNA and protein, as well as between protein and protein.
- Have the potential to quickly and reliably sequence the entire human genome for less than $1000, and possibly for even less than $100.
- The detection principle is based on monitoring the ionic current passing through the nanopore as a voltage is applied across the membrane.
- When the nanopore is of molecular dimensions, passage of molecules (e.g., DNA) cause interruptions of the current level, leading to a signal.
- Since DNA bases (adenine, thymine, cytosine and guanine) are different from each other in atomic scale, it is essential to collect base-specific information at atomic level to correspond the DNA sequence with the measured signals.
- According to the different types of the signals, the detection methods can be roughly classified into two categories:
    - The electrical detection methods and the optical readout method.
- In the 1990s, Church et al. And Deamer and Akeson separately proposed that it is possible to sequence DNA using nanopore sensors

Types of Nanopore sequencing:

- Nanopore technologies can be broadly divided into two categories:
- 1. Biological nanopore
- 2. Solid-state nanopore
- More recently, hybrid nanopores have been proposed to take advantage of the features of both biological and solid-state nanopores.

94. Biological Nanopore sequencing

- **Biological Nanopores**: Also called transmembrane protein channels
- Are usually inserted into a substrate, such as planar lipid bilayers, liposomes, or other polymer films
- Nanopores formed by pore-forming proteins
- Examples:

- alpha-hemolysin
- MspA
- phi 29

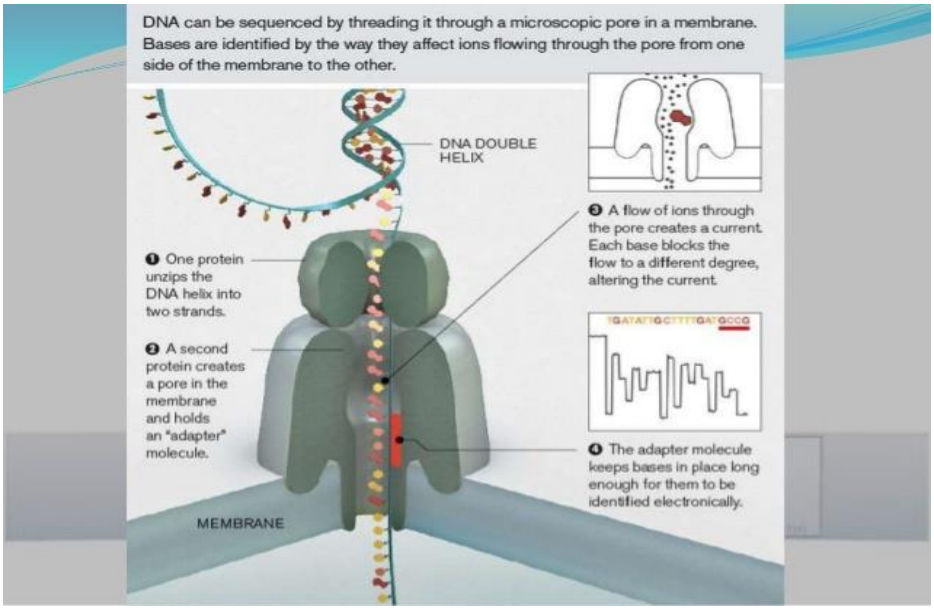
Alpha Hemolysin:

- a-Hemolysin (a-HL, also called a- toxin)
- a-HL is an exotoxin secreted by the bacterium Staphylococcus aureus, a human pathogen.
- This mushroom-shaped heptameric protein normally forms pores in the membranes.
- The limited pore size (1.4 nm) has restricted its application in the analysis of ssDNA, RNA, or small molecules.

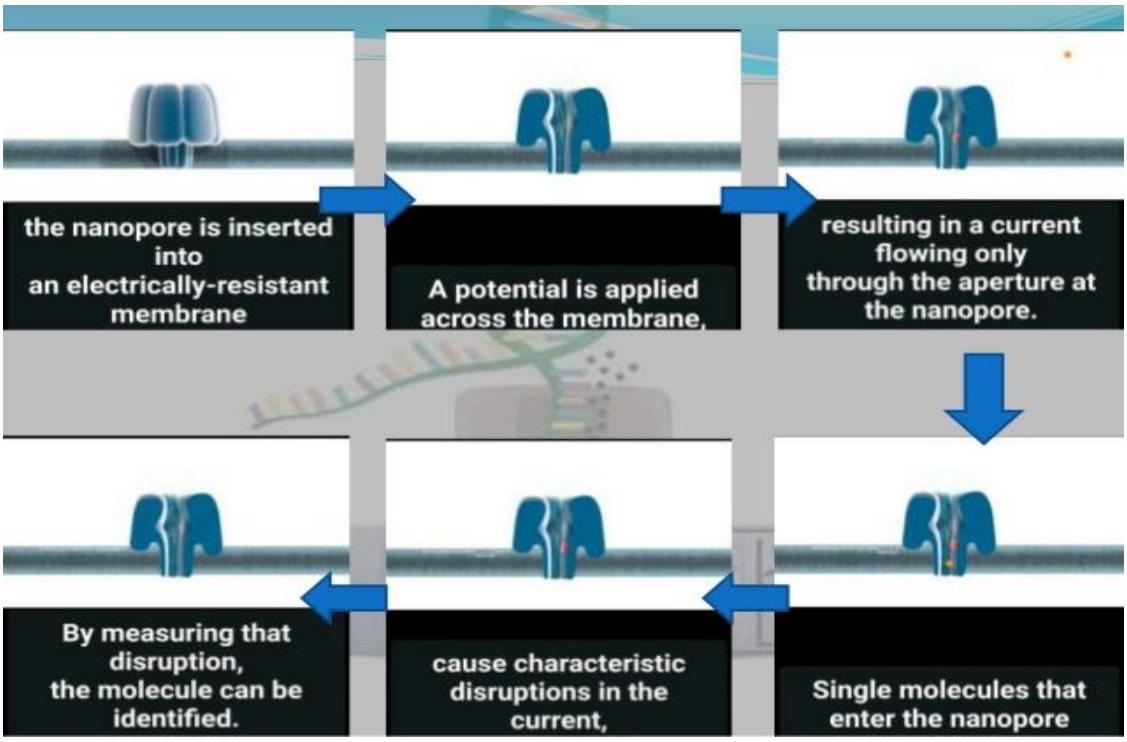
Mycobacterium smegmatis porin A (MspA)

- Powerful nanopore for reading information from four nucleotides simultaneously.
- By Butler et al. in 2008.
- The channel of the MspA octamer is 1 nm in diameter at the minimal point, which is relatively small and narrow, compared to that of alpha-HL.
- Thus, it can improve the spatial resolution of ssDNA sequencing

Principle:



Working Process:



Advantages:

- Their well-defined and highly-reproducible nanopore size and structure.
- Biological nanopores can be modified easily with modern molecular biology techniques, such as mutating the nucleotide sequence to change the amino acid residue at a specific site

Disadvantages:

- The environmental demands (like temperature, electrolyte concentration and Ph) of nanopores to keep their biological activities.
- The fragility of lipid bilayer makes the biological nanopore breakdown easily

95. Biopharmaceuticals I

- Biopharmaceuticals are defined as pharmaceuticals manufactured by biotechnology methods, with the products having biological sources, usually involving live organisms or their active components
- A protein or nucleic acid used as a drug, which has been produced by living organisms as a result of biotechnology

Includes products

- Monoclonal antibodies
- Recombinant proteins or oligonucleotides used as vaccines

- Nucleotide sequences and vectors used in gene therapy
- Wide range of recombinant proteins used as drugs.

Excluded products:

- Proteins and nucleic acid extracted from sources that have not been manipulated using biotechnology e.g.
- Recombinant human growth hormone
- Growth hormone extracted from pituitary gland.

Pharmaceutical Technology:

- The methods and techniques that involve the use of living organisms (such as cells, bacteria, yeast and others) are tools to perform specific industrial or manufacturing process are called biotechnology
- Pharmaceutical Technology will continue to provide new breakthroughs in medical research in the years to come, leading to treatment in field which have previously eluded us (including AIDS, cancer asthma, Parkinson's disease, Alzheimer disease)

The range of Biopharmaceuticals:

- First human recombinant protein somatostatin expressed in E. coli….1970s
- True biopharmaceutical produced was recombinant protein insulin…1982
- First therapeutic use of monoclonal antibody was in 1986.
- Hormones
- Clotting factors
- Cytokines
- Growth factors
- Thrombolytic factors and enzymes etc.
- Rapid increase in growth of biopharmaceutical products is due to combination of
  - Increase in our understanding of disease processes
  - The relative ease with which same technique can be used to produce biologically active molecules.

Advantages:

- BP can be engineered to be identical to human proteins e.g., insulin
- Molecularly engineered biopharmaceuticals avoid the need to extract human protein thus avoiding transmission of diseases.

- BP are easy to produce (in lab) and less amount is sufficient.
- High degree of specificity i.e., limited to a certain species

Disadvantages:

- Large scale production is expensive that's why these are expensive drugs.
- Generic products are as expensive as new conventional drugs.
- Risk of contamination from infectious agents, cells or products
- Very small changes can produce significant changes in the product.
- An ideal drug is one which is given orally and one dose is enough unlike BP products.
- Despite their extreme selectivity these drugs do produce side effects.

96. Biopharmaceuticals II

- Biopharmaceuticals are defined as pharmaceuticals manufactured by biotechnology methods, with the products having biological sources, usually involving live organisms or their active components
- A protein or nucleic acid used as a drug, which has been produced by living organisms as a result of biotechnology

Large Scale Production of Biopharmaceuticals:

- Most commercial production of biopharmaceuticals uses cultured mammalian cells. Other expression systems are
- Bacteria
- E. coli and Bacillus subtilis
- Simple physiology
- Grow rapidly
- Large product yields
- Made to excrete product into the medium to ease the recovery process.
- However not suitable for mammalian protein because of no glycosylation
- Insulin, growth hormone and some other are produce through bacterial system

Yeast:

- Widely used in BP industry
- Post translational modifications occurs (to some extent).

- Made to excrete product into the medium to ease the recovery process
- One disadvantage is the active production of proteases that destroys proteins which can be engineered.

Transgenic Animals:

- First transgenic animal was a mouse produced in yale 1980.
- 1996 Dolly was cloned
- Several companies are producing recombinant protein from transgenic sheep and goats from their milk.
- This technique has several advantages over other systems while making original transgenic animal is complex.
- Yield is comparable to large bioreactors.
- But still potential risk of animal disease transmission.
- Main drawback is the animal welfare.

Transgenic Plants:

- Number of therapeutic medicines are produced through tobacco and potato plants.
- Transgenic plants are much cheaper than animals.
- Product purification is costly in plants.
- Use of raw eating plants to avoid extraction and purification step.
- Concentration of products vary with different environment.
- Cross pollination and cross contamination cannot be avoided even with good manufacturing practices.

Mammalian cell Expression System:

- First used for the production of inactivated polio vaccine.
- Nowadays continuous immortalized cells lines are used instead of primary cell lines.
- General method is culturing transfected cells on large scale and then target protein is recovered.
- Practicing good manufacturing practices.
- Any small change require change in license product

Good Manufacturing Practices:

- Adherence to GMP is requirement of the licensing of any drug.

- All the components of manufacturing process must be strictly defined, to make sure that the product is made to the highest levels of safety and efficacy.
- Any change in the system i.e., even to increase the manufacturing product is likely to necessitate a change in the license.

Cell Characterization & Banking System:

- It is necessary to validate the nature and source of all cell lines in the production of BP.
- Several banks are maintained to ensure the safety and validity of this valuable resource. There would be:
  - Two master banks where original stocks are stored
  - One or more working banks, where cells are withdrawn for working.
  - Banked cells are continuously checked to ensure that they are of right type i.e. genetically stable and not infected or contaminated.
  - Age of cells are also monitored.

Product Characterization and validation:

- Careful characterization of the product is essential as small change can lead to change in product activity. A single amino acid change can affect activity to 50%.
- Cells used also effect the nature of product yielded.
- Changes in the conditions under which cells are cultured can affect the product yield to considerable extent.

97. Pharmacogenetics.

- Introduction:
- **Pharmacogenomics:** is the science of how genes affect the way people respond to drugs
- How genes affect, the way our body processes drugs (pharmacokinetics)
- The interaction of drugs with receptors (pharmacodynamics)
- The treatment efficacy and adverse side effects
- **Pharmacogenetics**
- A subset of ‘pharmacogenomics’
- The study of how inherited variation affects drug response and metabolism

Why is this a good Approach?

- Drugs can be dangerous

- Many people have severe adverse reactions to drugs
- Many people respond to drugs at different doses
- Many drug treatments are horribly unpleasant, painful
- Drugs are expensive (to take and to make)
- Ineffective drugs are a waste of money to take
- Drug development needs to account for response variability
- Genetics provide a prior information
- Genetics don't change (except in cancer)
- Genetics can point to the cause not just the symptom

Early Studies:

- Fast increase in awareness of the interaction of drug and drug response
- Many family studies, twin studies and 'top-down' genetic studies
- Very laborious experimental work through the study enzyme actions and clinical observation
- PCR sped things up a bit

CYP2D6

- 1975 Smith and colleagues ingest a drug they are testing
- He had a bad reaction but his colleagues did not
- Family studies revealed genetic inheritance
- Enzyme discovered and characterized
- Enzyme cDNA sequenced and variants found (1990)
- This family of gene important for many drugs

Genotyping:

- Uses a microarray to measure a limited predefined set of SNPs
- Very high throughput (fast)
- Very inexpensive (cheap)
- Excellent coverage of common variation (up to 5,000,000 SNPs)
- Relies on Linkage Disequilibrium

A T C G A A A T G C A T G A C C T T T G A T A T G A T C G G C T G C A G T C A G C

T T C G A A G T G C A T G A C T T T T G A C A T G A G C G G C G G C C C A C A G C

Common Variation Rare Variation No Recorded Variation

Deep Sequencing:

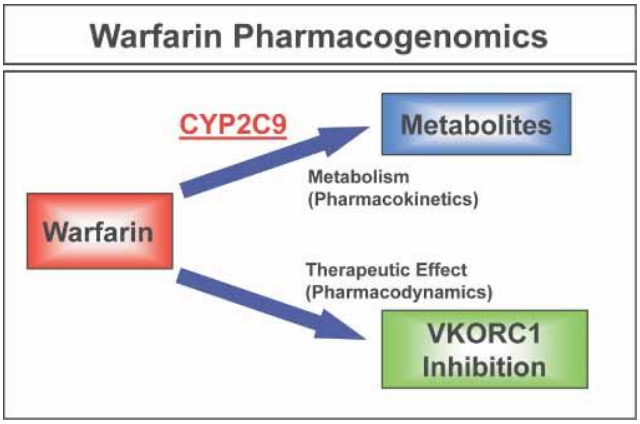
- Captures every base pair in the genome (3,000,000,000)
- (Currently) low throughput (slow)
- (Currently) Very expensive (> 10k)
- Captures common, rare, and personal variation
- New and hard to analyze

Utility of drugs:

- The utility of pharmacogenetics:
  - Determining appropriate dosing
  - Avoiding unnecessary toxic treatments
  - Ensuring maximal efficacy
  - Reducing adverse side effects
  - Developing or choosing novel treatments
  - Can also explain variable response to illicit drugs

Warfarin: A dosage Story:

- Most widely used anticoagulant in the world
- A “blood thinner”
- Prescribed doses vary widely (1-40mg / daily)
- Therapeutic index is very low
- High risk of bleeding early in treatment
- Two genes involved in metabolism: CYP2C9 and VKORC1



Plavix: A story about effectiveness

- Anti-clotting drug
- Prescribed for coronary artery disease and those who have suffered a heart attack or stroke or have a stent
- A "pro-drug"
- Converted to active form in the liver by CYP2C19

Pegasys: A toxic treatment story

- Pegylated Interferon α-2a
    - Interferons are proteins made in response to virus
- Treatment for Hepatitis B and C Virus
- Highly toxic treatment
- Highly variable response, especially in African Americans
- Very expensive

Personalized Medicine:

- There is an emerging goal among 'translational scientists' to make medical practice more personalized
- Pharmacogenetics is an important step towards that goal
- Personalized medicine is an emerging practice of medicine that uses an individual's genetic profile to guide decisions made in regard to the prevention, diagnosis, and treatment of disease.
- Knowledge of a patient's genetic profile can help doctors select the proper medication or therapy and administer it using the proper dose or regimen.

Concerns with this approach:

- How reliable are the tests?
- Are health care providers prepared to use this information?
- Will Insurance companies pay for the tests?
- Will tailor make medicine lead to discrimination?
- Will ethnic biases in science bias treatment developments?
- Will this affect people's privacy?

98: Principle of Drug Delivery System.

Drug Delivery:

- The appropriate administration of drugs through various routes in the body for the purpose of improving health
- It is highly interdisciplinary
- It is not a young field
- It has recently evolved to take into consideration

Oral administration:

- **Advantages**
  - Patient: Convenience, not invasive, higher compliance
  - Manufacture: well established processes, available infrastructure
- **Disadvantages**
  - Unconscious patients cannot take dose
  - Low solubility
  - Low permeability
  - Degradation by GI enzymes or flora
  - First pass metabolism
  - Food interactions
  - Irregular absorption
  - **Traditional oral delivery systems**
  - Tablets
  - Capsules
  - Soft gelatin capsules

- Suspensions
- Elixirs

Buccal/Sublingual:

- **Advantages**
  - By-pass First pass metabolism
  - Rapid absorption
  - Low enzymatic activity
- **Disadvantages**
  - Discomfort during dissolution
  - Probability of swallowing- loss of effect
  - Small doses

Rectal:

- **Advantages**
  - By-pass first pass metabolism
  - Useful for children
- **Disadvantages**
  - Absorption depends on disease state
  - Degradation by bacterial flora
  - Uncomfortable

Intravenous (IV):

- **Advantages**
  - Drug 100% bioavailable
  - Rapid response
  - Total control of blood concentration
  - Maximize incorporation of degradable drugs
- **Disadvantages**
  - Invasive
  - Trained personnel

    - Possible toxicity due to incorrect dosing
    - Sterility

Subcutaneous:

- **Advantages**
    - Patient self-administration
    - Slow, complete absorption
- **Disadvantages**
    - Invasive
    - Irritation, inflammation
    - Maximum dose volume - 2mL

Intramuscular:

- **Advantages**
    - Patient can administer the drug himself
    - Larger volume than subcutaneous
    - By-pass first pass metabolism
- **Disadvantages**
    - Invasive – patient discomfort
    - Irritation, inflammation
    - May require some training

Inhalers:

- **Advantages**
    - Gases are rapidly absorbed
    - Easy to use
- **Disadvantages**
    - Solids and liquids can be absorbed if size is below 0.5um

Factors Influencing the Selection of the Delivery Route

- pH
- pKa - ionization

- Temperature
- Concentration
- Crystallinity
- Particle size
- State of hydration

Factors Influencing Bioavailability

- Delivery route
- The site of measurement
- Type of animal employed
- Physiological state of the animal/human

99. Synthetic Pharmaceuticals.

Drug and Medicine:

- Drug is a chemical substance that interacts with a living system and produces a biological response
- **Drugs** are considered to be such things as stimulants, narcotics, and hallucinogens, and often are used in the medical field to treat or cure disease. **Medicine**, however, is a term for a drug when that drug is used to cure, mitigate, treat, or prevent a disease.

What criteria MUST new drugs meet?

- Drugs must address a new need or provide a significant "added benefit" over an existing medicine
- Drugs must also meet these criteria:
  - Must be safe
  - Effective
  - Of high quality
  - Cost effective

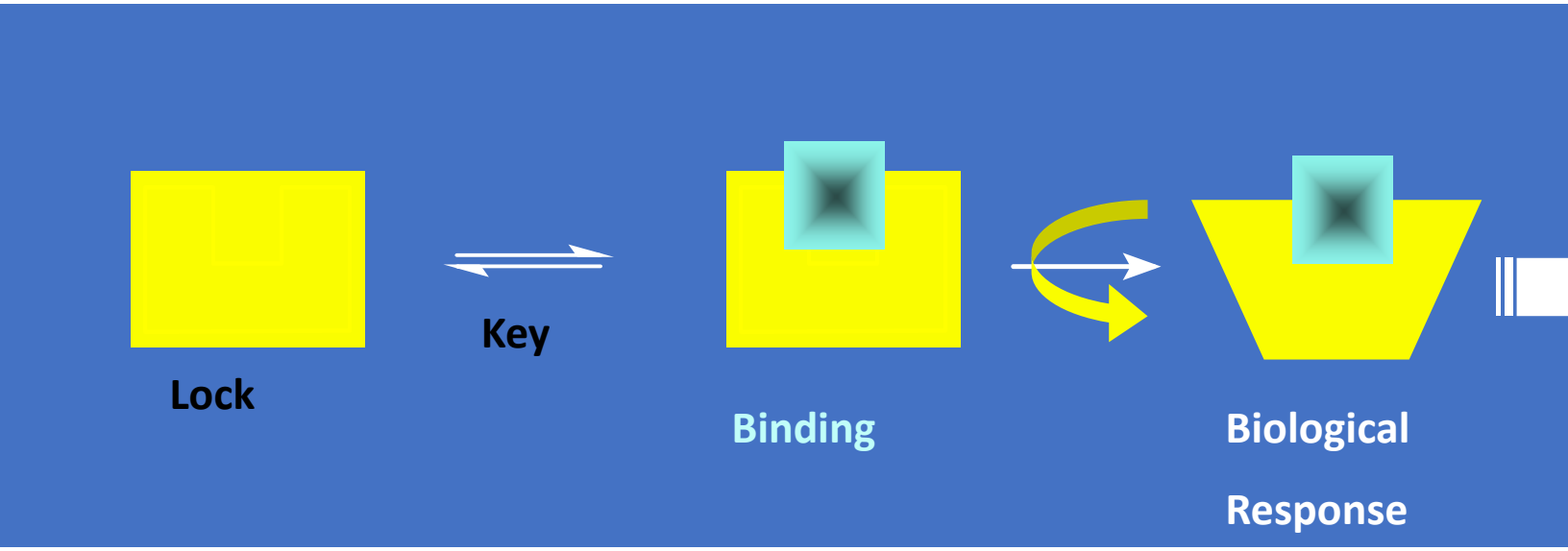
Major Therapeutic Targets

- **Infectious disease – anti-infectives**
  - Anti-bacterial, anti-viral, anti-parasitic drugs
- **Metabolic disease**

- cancer, cardiovascular, diabetes, inflammation, high blood pressure, neurological disease, pain

- **Other aspects of health care**
  - Hormonal treatments, contraception, vaccines, immunosuppressant, anaesthetics, nutraceuticals,

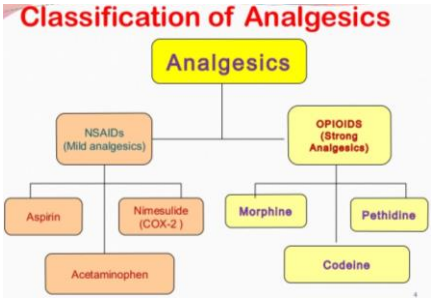
The "Lock and Key" analogy



- Here the KEY is the natural substrate
- Binding of the KEY to the LOCK (an enzyme or a receptor) then causes a response – a shape change in the protein/receptor

Analgesics:

- Analgesic drugs act in various ways on the peripheral and central nervous systems.
- They are distinct from anesthetics, which reversibly eliminate sensation.



Paracetamol and NSAIDs

- The exact mechanism of action of paracetamol/acetaminophen is uncertain but appears to act centrally in the brain rather than peripherally in nerve endings. Aspirin and the other non-steroidal anti-inflammatory drugs (NSAIDs) inhibit cyclooxygenases, leading to a decrease in prostaglandin production. In contrast to paracetamol and the opioids, this reduces not only pain but inflammation as well.

Opioids

- Morphine, the archetypal opioid, all exert a similar influence on the cerebral opioid receptor system.
- Opioids, while very effective analgesics, may have some unpleasant side-effects. Patients starting morphine may experience nausea and vomiting, itching, constipation etc.
- When used appropriately, opioids and similar narcotic analgesics are otherwise safe and effective, however risks such as addiction and the body's becoming used to the drug (tolerance) can occur.
- **The effect of tolerance means that frequent use of the drug may result in its diminished effect so, when safe to do so, the dosage may need to be increased to maintain effectiveness**

Anti-inflammatory

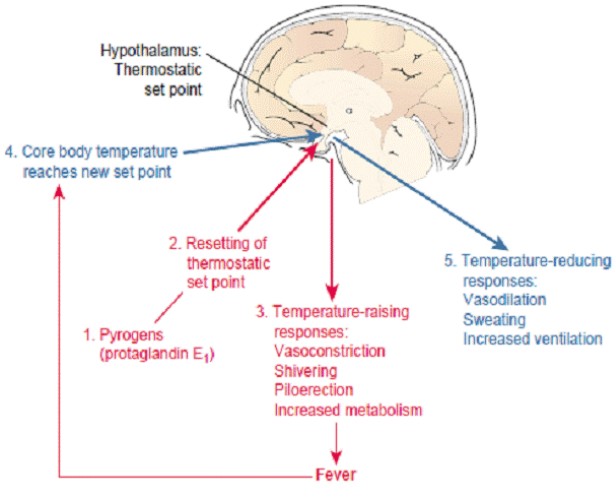
- Anti-inflammatory refers to the property of a substance or treatment that reduces inflammation or swelling.
- COX-2 inhibitors
- Prostaglandins inhibitors

Inflammation

Cardinal Signs	Physiological rationale
1. Rubor (Redness)	Increased Blood flow
2. Tumor (Swelling)	Exudation of fluid
3. Calor (Heat)	Increased Blood flow , Exudation of fluid, Release of inflammatory mediators
4. Dolor (Pain)	Stretching of pain receptors and nerves by inflammatory exudates , chemical mediators
5. Functio laesa (Loss of function)	Pain, Disruption of tissue structure, Fibroplasia and metaplasia

Antipyretic

- Antipyretics cause the hypothalamus to override an interleukin-induced increase in temperature.



Antiemetic

- Antiemetic is a drug that is effective against vomiting and nausea.
- Block serotonin receptors in the central nervous system and gastrointestinal tract

Antihistamine

- Antagonists of the histamine H_1-receptor are used to treat allergic reactions.
- Histamine produces increased vascular permeability, causing fluid to escape from capillaries into tissues, which leads to the classic symptoms of an allergic reaction — a runny nose and watery eyes.

Steroid (Life Saving Drug)

- Steroids reduce the production of chemicals that cause inflammation. This helps keep tissue damage as low as possible. Steroids also reduce the activity of the immune system by affecting the way white blood cells work.
- Given when,
    - Body defense is poor
    - Extensive tissue damage

Anesthetics

- An **anesthetic** is a drug that causes anesthesia, which is a reversible loss of sensation.

1. **General anesthetics**
2. **Local anesthetics**

Antibiotics

- Medications used to treat bacterial infections

1. Sulfonamides
2. Penicillin
3. Cephalosporins
4. Tetracyclines
5. Aminoglycosides
6. Quinolones
7. Macrolides

How Antibiotics Work

- Block protein formation
    - Macrolides, Tetracyclines, Aminoglycosides
- Inhibit cell wall formation
    - Pencil line, Cephalosporins
- Interfere with DNA formation
    - Nalidixic acid (synthetic quinolone), Quinolone
- Prevent folic acid synthesis
    - Sulfonamides

Others:

Diuretics

Anti- depressants

Contraceptives

Anti-hypertensive

Athletic steroids

Anti-coagulants

100. Drug Abuse.

What are drugs and What is their normal use?

- A chemical substance that is used in treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being.

- They altering biochemical and physiological process of tissue or organism.

Street Drug

- Drug that are taken for non-medical reasons (usually for mind altering effects)
- Can lead to
  - Physical damage
  - Mental damage
  - Dependence and addiction
- Include alcohol, heroin, methamphetamine, crack, cocaine and marijuana (Cannabis)

Drug Abuse

- A form of substance-related disorder
- Patterned use of a drug in which the user consumes the substance in amounts or with methods which are harmful to themselves or others
- Drugs most often associated with this term include:
  - Alcohol, substituted amphetamines, barbiturates, benzodiazepines, cocaine, methaqualone, and opioids.

Drug Addiction

- Chronic, often relapsing brain disease that causes compulsive drug seeking and use, despite harmful consequences to the addicted individual and to those around drug addicts.

Difference between drug abuse and addiction

- **Drug abuse** refers to using drugs even though it has become a problem in life.
- **Drug addiction** refers to the substance, taken in higher amounts or taken over a longer period of time than the user intended.

Cause of Substance Abuse

- Not clear, with theories including:
  - A genetic disposition
  - Learned from others
  - A habit which if addiction develops, manifests as a chronic debilitating disease

Genetics

- Factors that make it harder to become addicted also may be genetic.
- Example: an individual may feel sick from a drug that makes other people feel good.

- There is no single addiction gene but many interacting genes.
- Social and environmental factor contribute to risk of addiction

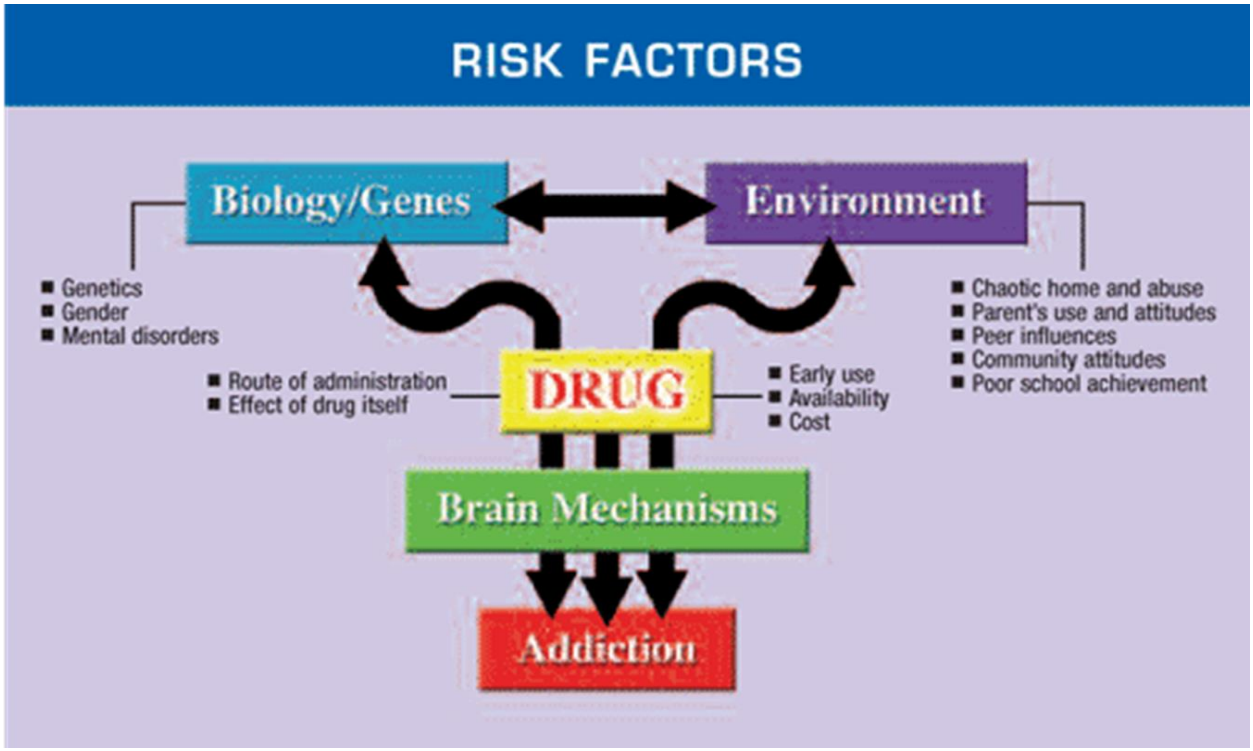
Other Factors

- 1. **Gender**
- Men are twice as likely to have problems with drugs
- 2. **DRUG**
- Cause temporary, pleasurable feelings
- Dependent
- 3. **Mental condition**
- Anxiety, depression and other mental conditions

Individual factors	Environmental factors	Constitutional factors
Personality	Socioeconomic status	Biological/genetic predisposition to use/misuse drugs
Gender	Poverty	
Anxiety	Delinquency	
Stress	Family background/disturbance	
Power needs	Peer pressure	
Age	Ideology/religion	
Intelligence	Educational opportunities	
Psychological health	Educational disturbance	
Life events	Truancy	
Predisposition to take risks	Drug availability	
Hedonism	Drug price	
Self-destructiveness	Unemployment/job opportunities	
Curiosity	Anomie (lack of social/moral standards)	
	Alienation (rebelliousness)	
	Tradition	
	Legal arrangements	
	Historical factors	

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Risk Factors:



Conclusions:

- Just because you are prone to addiction doesn't mean you're going to become addicted. It just means you've got to be careful.
- There are many ways that genes could cause one person to be more vulnerable to addiction than another.